Lithium, Tech and Uranium : A Quantitative Dive into ETF Time Series

Author : Gianni MARCHETTI

Master in Financial Modelling

July 8, 2025

Contents

1 Introduction

Exchange-Traded Funds (ETFs) are investment funds that are traded on stock exchanges, much like individual stocks. Each ETF typically tracks a specific index, sector, commodity, or asset class, offering investors a low-cost and transparent way to gain diversified exposure. Over the past decade, ETFs have become increasingly popular among both institutional and retail investors due to their flexibility, liquidity, and efficiency.

As of recent statistics, the ETF market has grown significantly. In the United States, there is over \$5.4 trillion invested in equity ETFs and \$1.4 trillion in fixed-income ETFs. In Europe, the amounts are approximately \$1.0 trillion in equity ETFs and \$0.4 trillion in fixed-income ETFs, while Asia has around \$0.9 trillion in equity ETFs and \$0.1 trillion in fixed-income ETFs. In the first quarter of 2023, ETFs accounted for 32% of total dollar volume traded in U.S. stock markets, 11% in Europe, and 13% in Asia (*Global ETF Market Facts*, BlackRock).

This project focuses on three original and sector-specific ETFs: **KWEB** (Chinese internet and technology), **LIT** (lithium and battery technology), and **URA** (uranium and nuclear energy). Each of these ETFs represents a distinct macroeconomic and industrial theme. They were selected based on their global relevance, relatively low correlation, and a trading history of at least seven years, which allows for robust time series analysis.

The aim of this project is to conduct a comprehensive empirical study of these ETFs using weekly data. We begin by exploring the price dynamics and computing descriptive statistics of the log-returns. Then, we assess distributional properties through normality testing, examine the correlation structure and autocorrelation patterns. Following this, we model the log-prices using ARIMA(6,1,0), and apply stationarity tests such as Augmented Dickey-Fuller (ADF) and KPSS. Finally, we investigate potential long-term equilibrium relationships using the Engle-Granger cointegration approach.

The entire workflow is implemented in Python and available on my GitHub profile, with a strong focus on clean data visualization and interpretation, in line with modern empirical finance practices.

2 Data Description and Price Dynamics

This section provides a detailed overview of the three selected ETFs, focusing on both their fundamental characteristics and their price behavior over time. We begin by introducing the thematic focus and composition of each ETF, highlighting their economic relevance and sectoral positioning.

Next, we analyze the historical evolution of their prices using normalized series to enable cross-comparison, independent of initial price levels. These elements lay the foundation for the subsequent statistical analysis and time series modeling conducted in later sections.

2.1 ETF Characteristics

KWEB (KraneShares CSI China Internet ETF) is a thematic equity ETF that offers exposure to Chinese internet and technology companies. It seeks to track the CSI Overseas China Internet Index, which includes companies that develop and market internet software and services, provide internet retail, or operate mobile applications and social media platforms.

The ETF is issued by KraneShares, an asset management firm based in New York that specializes in China-focused exchange-traded funds. KraneShares aims to provide global investors with innovative ways to capture China's economic expansion and its increasing role in the global economy. KWEB is one of its flagship products and has attracted significant attention due to its targeted exposure to Chinese internet giants.

KWEB includes major companies such as Alibaba, Tencent, Meituan, JD.com, Baidu, and Pinduoduo, many of which are listed in the United States via American Depositary Receipts (ADRs) or have dual listings in Hong Kong. The ETF is heavily weighted toward e-commerce, online media, cloud computing, and fintech, making it highly sensitive to regulatory changes and macroeconomic developments in China.

Due to its focus, KWEB is known for its high volatility and growth potential, offering an opportunity to invest in the digital transformation of China's consumer economy. The ETF was launched in 2013 and has since become a benchmark for tracking the performance of Chinese internet equities on the international stage.

The second tracker is **URA** (Global X Uranium ETF) is a thematic equity ETF designed to provide exposure to companies involved in the uranium industry, including mining, exploration, equipment, engineering, and the production of nuclear components. The fund seeks to track the Solactive Global Uranium & Nuclear Components Total Return Index, giving investors targeted access to a key component of the global energy mix.

The ETF is issued by Global X, a New York-based ETF provider that specializes in niche and forward-looking investment strategies. URA is part of the firm's suite of energy transition ETFs, which aim to capture long-term structural changes in how energy is produced and consumed globally.

URA's portfolio includes leading uranium producers and suppliers such as Cameco Corporation, Kazatomprom, NexGen Energy, and Denison Mines. It also holds companies involved in nuclear energy infrastructure and related technologies. The fund's exposure is globally diversified, with major holdings in Canada, Kazakhstan, Australia, and the United States.

Since its launch in 2010, URA has gained renewed investor interest amid the global push for decarbonization and energy security. Nuclear power is increasingly seen as a low-carbon, reliable alternative to fossil fuels. As such, URA offers investors a focused opportunity to participate in the long-term resurgence of nuclear energy, though it remains subject to volatility due to regulatory, political, and commodity price risks.

The last one is **LIT** (Global X Lithium & Battery Tech ETF) is a thematic ETF that provides exposure to the full lithium cycle, including mining, refining, and battery production. The fund seeks to track the Solactive Global Lithium Index, offering investors access to companies that are engaged in lithium mining or the manufacturing of lithium-based batteries used in electric vehicles (EVs), energy storage systems, smartphones, and other devices.

The ETF is also issued by Global X and LIT's holdings span the entire lithium value chain and include major companies such as Albemarle, SQM, Ganfeng Lithium, LG Chem, and Tesla. The fund's geographical allocation is diversified across the United States, China, South Korea, Chile, and Australia. It combines both upstream (extraction and refining) and downstream (battery production and integration) exposure.

Since its inception in 2010, LIT has attracted increasing interest from investors looking to capitalize on the transition toward clean energy and electrification. Its performance is closely linked to lithium prices and the expansion of the EV market, making it a high-growth yet high-volatility ETF aligned with long-term sustainability themes.

The data used to analyze these ETFs were retrieved from Yahoo Finance, with a weekly frequency, covering the period from January 1, 2015 to January 1, 2025.

2.2 Normalized Price Evolution

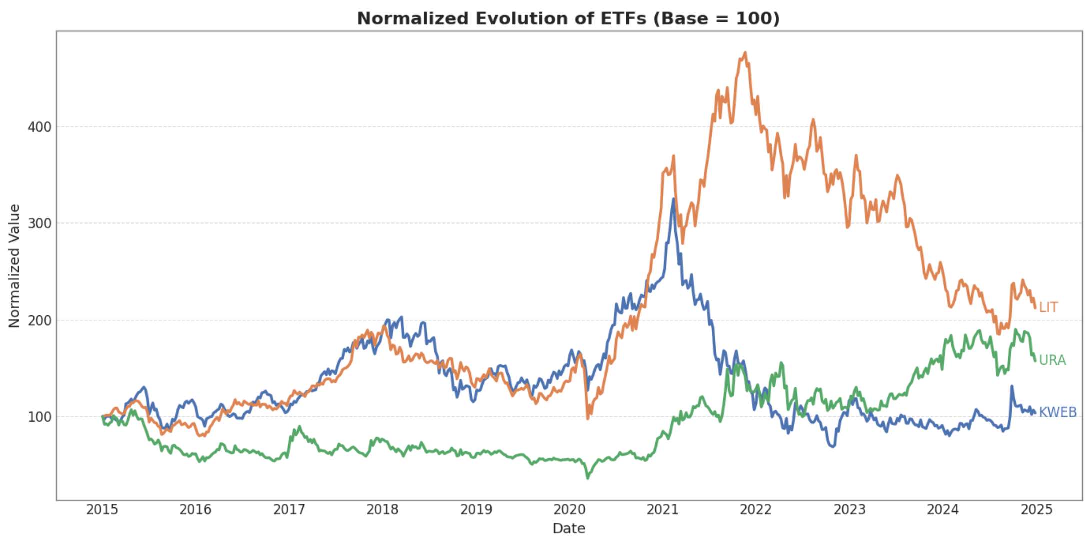

Figure 1: Normalized price evolution of KWEB, LIT, and URA (base 100)

Normalizing the prices to a base value of 100 at the beginning of the observation period allows for a direct comparison of the relative performance of each ETF over time, regardless of their absolute price levels. This method highlights percentage changes and facilitates a visual interpretation of growth dynamics.

As shown in Figure 1, the three ETFs exhibited very different trajectories between 2015 and 2025. LIT demonstrated the strongest upward momentum, particularly between 2020 and late 2021, where it peaked above 450. This surge reflects the explosive growth of the electric vehicle and battery technology sectors, driven by both investor enthusiasm and structural shifts toward clean energy.

KWEB, which represents Chinese internet and technology firms, experienced a significant rise between 2020 and early 2021, followed by a sharp and persistent decline. This trend aligns with the Chinese government's tightening regulatory environment on big tech and increased geopolitical uncertainty, especially affecting companies listed overseas.

URA, the ETF focusing on uranium and nuclear energy, remained relatively flat during the early part of the period. However, since 2020, it has shown steady and robust growth, surpassing KWEB in performance. This reflects renewed global interest in nuclear energy as a low-carbon alternative in the context of energy transition and rising fossil fuel prices.

Overall, LIT was the top performer over the entire period, albeit with high volatility. URA emerged as a consistent late-stage performer, while KWEB underperformed, especially in the post-COVID regulatory context.

3 Return Analysis and Statistical Properties

In this section, we analyze the statistical behavior of the weekly log returns of the three selected ETFs. The goal is to gain insights into their return distributions, risk characteristics, and dependencies. We begin with standard descriptive statistics, followed by tests for normality and distributional shape. We then examine the correlations between the ETFs, and finally explore their autocorrelation structures to detect potential time dependencies in returns.

3.1 Descriptive Statistics

	mean	std	skewness	kurtosis	max	min
Ticker						
KWEB	0.000067	0.050637	0.170412	2.640222	0.270973	-0.171936
LIT	0.001442	0.040340	-0.600189	4.086104	0.153490	-0.262523
URA	0.000870	0.052691	-0.007033	2.048535	0.200709	-0.237428

Figure 2: Descriptive statistics of weekly log returns for KWEB, LIT, and URA

From the descriptive statistics presented in Figure 2, we observe that LIT had the highest average weekly return (0.001 442), followed by URA (0.000 870) and KWEB (0.000 067). However, this higher return for LIT comes with a trade-off in the form of greater tail risk, as evidenced by its kurtosis of 4.09, the highest among the three ETFs.

In terms of volatility, KWEB and URA show similar standard deviations (0.050 637 and 0.052 691 respectively), both higher than LIT's (0.040 340). This indicates that KWEB and URA are subject to larger price swings week-to-week.

Skewness values reveal asymmetries in the return distributions. LIT is negatively skewed (-0.60), suggesting a greater likelihood of large negative returns. In contrast, KWEB has a slight positive skew (0.17), while URA is almost symmetrical (-0.007).

Finally, the kurtosis values for all three ETFs are above 2, indicating heavier tails than a normal distribution. LIT, in particular, shows strong leptokurtic behavior, pointing to a higher probability of extreme returns (both gains and losses).

These insights highlight that while LIT offered the highest average return over the period, it also exhibited the most pronounced tail risk and downside asymmetry.

3.2 Distribution and Normality

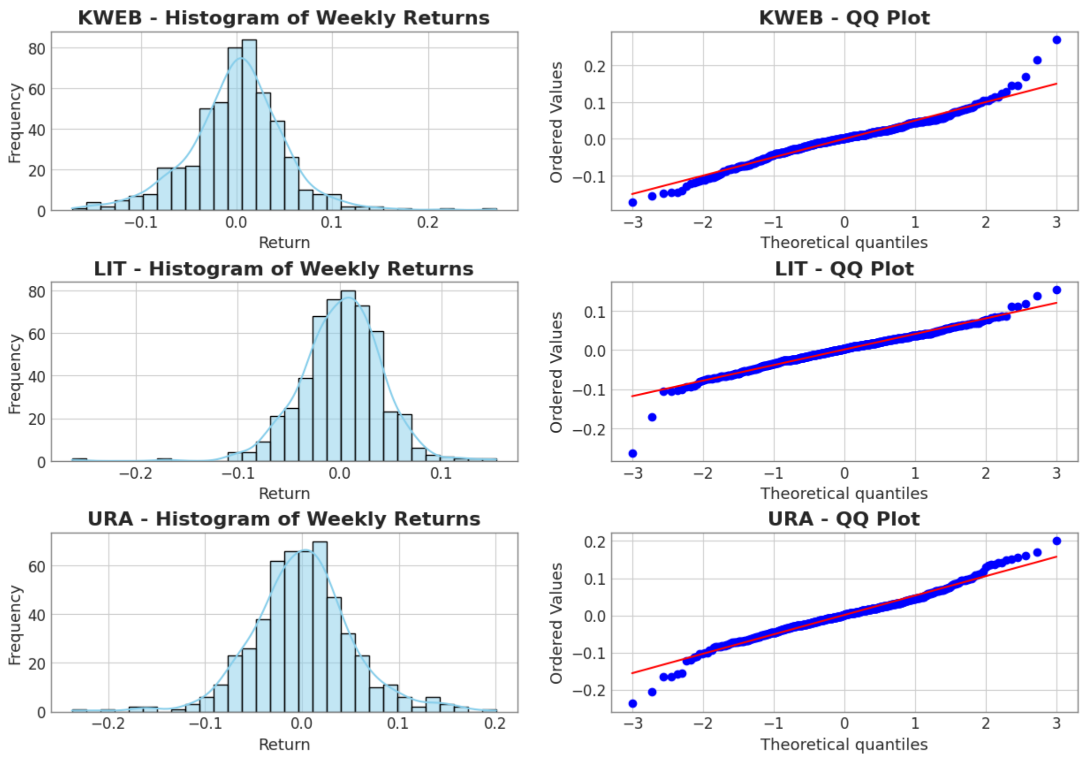

Figure 3: Histograms and QQ-plots of weekly log returns for KWEB, LIT, and URA

Figure 3 presents histograms with kernel density estimations alongside QQ-plots for the weekly log returns of KWEB, LIT, and URA. These visual tools allow us to assess the distributional properties of returns and evaluate the assumption of normality.

Starting with KWEB, the histogram reveals a moderately asymmetric distribution with a longer right tail. This aligns with its positive skewness (0.170) observed in Table 2. The QQ-plot for KWEB shows moderate deviations from the red theoretical line, particularly in the lower and upper quantiles, suggesting the presence of extreme values. The kurtosis of 2.64 supports this observation, indicating fatter tails than a normal distribution but not excessively so.

LIT displays a more symmetric histogram at first glance, but the left tail is visibly heavier. This observation is confirmed by the significant negative skewness (-0.60), pointing to a higher frequency of large negative returns. The QQ-plot illustrates clear departures from normality in both tails, especially on the left side, which confirms the asymmetry. Moreover, the high kurtosis of 4.09 reflects the leptokurtic nature of LIT's returns — that is, the distribution has a sharp peak and fat tails, increasing the likelihood of extreme outcomes.

For URA, the histogram appears more balanced and bell-shaped. The skewness is nearly zero (-0.007), suggesting a symmetric distribution. However, the QQ-plot still exhibits visible deviations from the normal line in both tails, indicating that URA's return distribution, while more symmetric, still contains extreme values. The kurtosis value of 2.05, though lower than LIT's, confirms that the distribution remains leptokurtic relative to the normal benchmark (kurtosis $= 3$ in standard terms, or 0 in excess kurtosis).

In summary, all three ETFs exhibit departures from normality, as evidenced by the skewness, kurtosis, and visual diagnostics. LIT presents the most extreme characteristics with a heavy left tail and high peak, while URA is the closest to normal but still exhibits fat tails. These distributional features are critical for risk management and modeling, as they affect the likelihood of tail events not captured by standard Gaussian assumptions.

3.3 Correlation Matrix

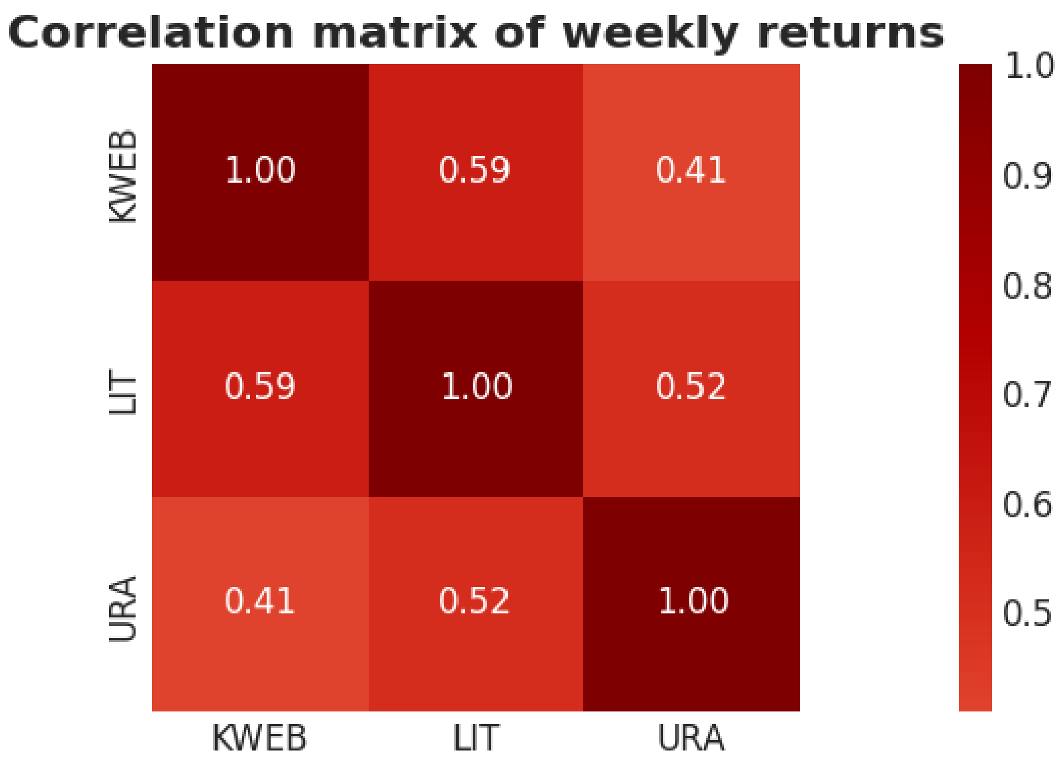

Figure 4: Correlation Matrix

The highest correlation is observed between KWEB and LIT (0.59), which may reflect periods of simultaneous risk-on or risk-off behavior among growth-oriented sectors. Both ETFs are influenced by investor sentiment toward innovation, tech-driven themes, and global macroeconomic conditions, despite their geographic and sectoral differences.

LIT and URA exhibit a moderate correlation (0.52), possibly due to their shared sensitivity to energy and environmental policy. Both sectors have benefited from global decarbonization trends, although they operate in distinct parts of the clean energy ecosystem (storage vs. production).

The weakest correlation is between KWEB and URA (0.41), which is expected given their different economic exposures (Chinese internet regulation versus global nuclear energy) and the relatively independent drivers of their performance.

These results suggest that while there is some co-movement among the ETFs, they retain a reasonable degree of diversification, making them suitable candidates for multivariate time series modeling or portfolio construction.

3.4 Autocorrelation (ACF)

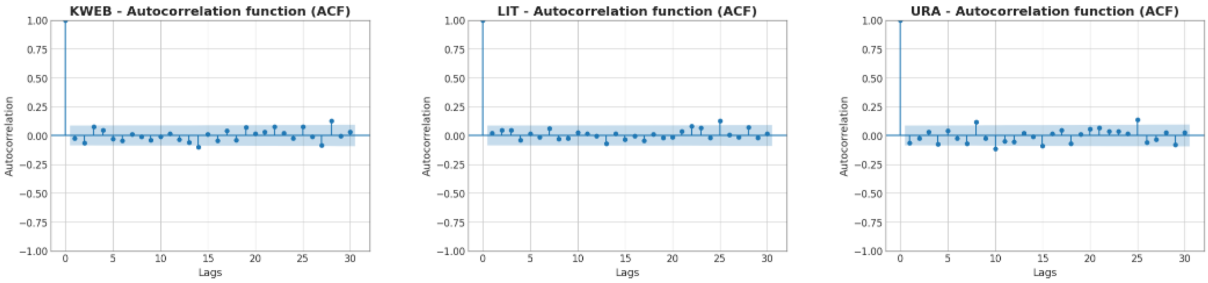

Figure 5: Autocorrelation functions (ACF) of weekly log returns for KWEB, LIT, and URA

The autocorrelation function (ACF) measures the linear correlation between a time series and its own past values across various lags. In the context of financial returns, it is used to assess whether past returns can help predict future returns. Significant autocorrelations indicate memory or structure in the time series, which could be exploited for forecasting or trading strategies.

Under the weak-form Efficient Market Hypothesis (EMH), asset returns should be unpredictable and resemble a white noise process, that is, their autocorrelations should be statistically insignificant. A strong ACF structure would contradict EMH and suggest potential predictability.

This shows the ACF of the weekly log returns for KWEB, LIT, and URA up to lag 30. For each ETF, the spike at lag 0 corresponds to the self-correlation, which is always equal to 1. The shaded bands represent the 95% confidence interval : autocorrelations that fall within this band are not statistically different from zero.

KWEB displays several small autocorrelation spikes around lags 5, 20, and 27, but all remain within the confidence band. This suggests the absence of significant linear dependence in the return series, consistent with market efficiency.

LIT shows a similar pattern, with minor fluctuations near zero and no clear temporal structure. Its return series behaves like a weakly stationary process with no memory, in line with EMH expectations.

URA exhibits slightly more persistent deviations at some lags, notably around lags 10 and 24. While these values approach the threshold of significance, they remain statistically insignificant. The return series appears weakly autocorrelated at best.

In summary, none of the ETFs exhibit significant autocorrelation in their weekly returns. This suggests that past return values do not contain useful linear information to predict future returns. Nevertheless, this conclusion is limited to linear dynamics; nonlinear patterns or volatility clustering may still be present and are better captured through further modeling techniques.

4 Time Series Modeling and Cointegration

This section applies time series econometric techniques to model the logarithmic prices of the selected ETFs and to assess the presence of long-term equilibrium relationships.

We begin by estimating an ARIMA(6,1,0) model, which captures short-term dynamics in the log-price series through autoregressive structure and differencing for stationarity. We then conduct stationarity testing using the Augmented Dickey-Fuller (ADF) and KPSS tests to formally evaluate the stochastic properties of the time series. Finally, we investigate potential cointegration between two ETFs using the Engle-Granger methodology to determine whether a stable long-term relationship exists despite short-term fluctuations.

This analysis is crucial for distinguishing between persistent trends, mean-reverting behavior, and co-movement patterns in financial time series.

For this analysis, only two of the three ETFs, **LIT** and **KWEB**, were selected for time series modeling and cointegration testing, to ensure interpretability and to focus on two economically distinct sectors.

4.1 ARIMA(6,1,0) Modeling

To model the weekly log-price dynamics of the selected ETFs, we apply an ARIMA(6,1,0) model. ARIMA stands for AutoRegressive Integrated Moving Average, and in this case, the specification includes six autoregressive lags (AR), one order of differencing (I), and no moving average term (MA). The differencing is applied to make the series stationary in mean.

Each estimated coefficient is associated with a *p-value*, which allows us to test the null hypothesis that the true coefficient is equal to zero. We adopt a conventional significance level of $\alpha = 0.05$. A p-value below this threshold indicates that the coefficient is statistically significant, meaning it likely has a real impact on the dependent variable. Conversely, a p-value above α suggests that the observed effect could be due to random chance and is not statistically meaningful.

The following table presents the estimated coefficients and associated statistics for the ARIMA(6,1,0) model fitted to the KWEB ETF.

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                   KWEB   No. Observations:                  522
Model:                 ARIMA(6, 1, 0)   Log Likelihood                 819.355
Date:                Sat, 05 Jul 2025   AIC                          -1624.711
Time:                        15:23:03   BIC                          -1594.920
Sample:                    01-01-2015   HQIC                         -1613.042
                         - 12-26-2024
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.0261      0.037     -0.712      0.476      -0.098       0.046
ar.L2         -0.0563      0.037     -1.513      0.130      -0.129       0.017
ar.L3          0.0795      0.043      1.865      0.062      -0.004       0.163
ar.L4          0.0427      0.040      1.075      0.283      -0.035       0.120
ar.L5         -0.0184      0.038     -0.488      0.625      -0.092       0.055
ar.L6         -0.0453      0.047     -0.954      0.340      -0.138       0.048
sigma2         0.0025      0.000     22.565      0.000       0.002       0.003
```

Figure 6: Estimated ARIMA(6,1,0) coefficients for KWEB (weekly log-prices)

Among the six autoregressive lags, none of the coefficients are statistically significant at the 5% level. The lowest p-value corresponds to the third lag (`ar.L3`), with a value of 0.062, which remains above the chosen threshold of $\alpha = 0.05$. As a result, we cannot conclude that any of the lagged returns have a statistically meaningful effect on the current log-price change. The estimated coefficients are small in magnitude and alternate in sign, suggesting noise rather than structured temporal dependence.

This result supports the hypothesis that KWEB's weekly log-returns behave in a weakly random fashion, consistent with the weak-form Efficient Market Hypothesis. While `ar.L3` shows a slight indication of delayed autocorrelation, it lacks the statistical strength required to reject the null hypothesis of no effect.

Overall, the model captures some basic structure but confirms that past weekly returns offer little predictive power for current returns in KWEB.

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                    LIT   No. Observations:                  522
Model:                 ARIMA(6, 1, 0)   Log Likelihood                 935.391
Date:                Sat, 05 Jul 2025   AIC                          -1856.781
Time:                        15:23:03   BIC                          -1826.991
Sample:                    01-01-2015   HQIC                         -1845.112
                         - 12-26-2024
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.0202      0.029      0.703      0.482      -0.036       0.076
ar.L2          0.0485      0.033      1.473      0.141      -0.016       0.113
ar.L3          0.0478      0.036      1.332      0.183      -0.023       0.118
ar.L4         -0.0421      0.044     -0.956      0.339      -0.128       0.044
ar.L5          0.0151      0.048      0.317      0.751      -0.078       0.108
ar.L6         -0.0130      0.038     -0.346      0.730      -0.087       0.061
sigma2         0.0016   6.81e-05     23.700      0.000       0.001       0.002
```

Figure 7: Estimated ARIMA(6,1,0) coefficients for LIT (weekly log-prices)

The ARIMA(6,1,0) model was fitted to the log-price series of LIT to examine potential short-term dynamics. As with KWEB, six autoregressive lags were included following first differencing to achieve stationarity.

The estimation results indicate that none of the lag coefficients (`ar.L1` to `ar.L6`) are statistically significant at the 5% level. All p-values are well above the conventional threshold (`p > 0.05`), suggesting that past weekly returns do not provide any meaningful explanatory power for current changes in LIT's log-price.

The signs and magnitudes of the estimated coefficients vary without clear structure, and their standard errors are relatively large compared to the coefficients themselves. This further supports the conclusion that LIT's weekly log-returns behave like a white noise process.

Overall, the lack of statistically significant autoregressive components reinforces the idea that LIT follows a largely unpredictable path in the short term, consistent with the weak-form Efficient Market Hypothesis.

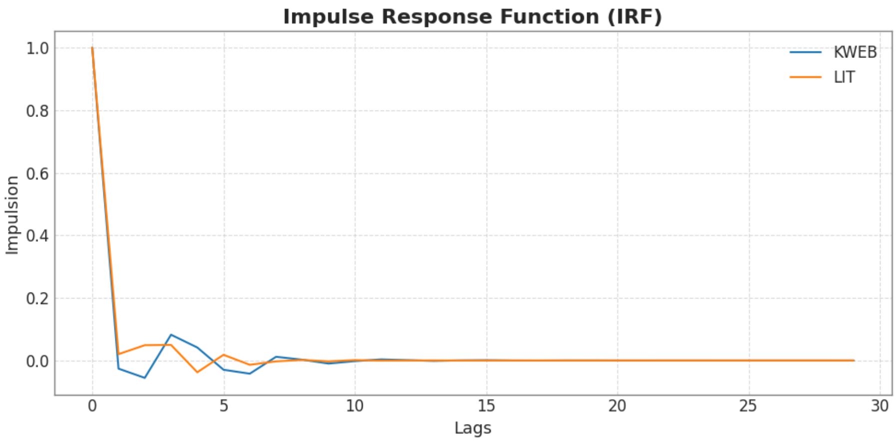

Figure 8: Impulse Response Function (IRF) for KWEB and LIT based on ARIMA(6,1,0)

The Impulse Response Function (IRF) is used to trace the dynamic effect of a one-time shock on the current and future values of a time series. In the context of ARIMA models, it allows us to observe how a one-unit innovation affects the system over time, helping to reveal short-term memory or persistence.

Figure 8 displays the IRFs for **KWEB** and **LIT**, derived from their respective ARIMA(6,1,0) models. Both ETFs exhibit a strong and immediate response to the initial impulse at lag 0, which is expected, as the shock directly affects the current observation.

Following the initial impact, the responses for both series rapidly decay toward zero within the first 5 lags. This fast decay indicates that the effect of a shock is transitory and does not persist in the medium or long term. The slight oscillations observed in KWEB's response (lags 2–5) suggest minor short-run interactions in the autoregressive process, consistent with the mild structure captured in its ARIMA estimation.

LIT, on the other hand, shows an even quicker stabilization, with negligible movement beyond the first lag. This aligns with the previously observed lack of significant autoregressive coefficients in the ARIMA model, reinforcing the conclusion that LIT's log-price changes are largely unpredictable and revert quickly after shocks.

Overall, the IRFs confirm that both ETFs have low persistence in weekly returns, and that shocks to either series tend to dissipate rapidly over time.

4.2 Stationarity Testing (ADF and KPSS)

Before applying models that assume stable statistical properties over time, it is essential to verify whether the underlying time series are stationary. A stationary time series has a constant mean, variance, and autocorrelation structure over time. Non-stationary data may lead to spurious regressions and unreliable forecasts.

To formally assess stationarity, we employ two complementary tests: the Augmented Dickey-Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. These tests differ in their hypotheses and jointly provide a more complete picture of the time series' behavior.

The ADF test has the null hypothesis (H_0) that the series contains a unit root—that is, it is non-stationary, versus the alternative hypothesis (H_1) that the series is stationary.

Conversely, the KPSS test takes the opposite stance: its null hypothesis (H_0) assumes that the series is stationary around a deterministic trend (trend-stationary), while the alternative hypothesis (H_1) is that the series is non-stationary due to a unit root.

Interpreting both tests together provides more robust conclusions. For example, if the ADF fails to reject non-stationarity and the KPSS rejects stationarity, we gain strong evidence that the series is indeed non-stationary.

We applied both the Augmented Dickey-Fuller (ADF) and KPSS tests to the weekly log-price series of KWEB and LIT in order to assess their stationarity properties.

For **KWEB**, the ADF test produces a p-value of 0.3282, and for **LIT**, a p-value of 0.6017. In both cases, these values are well above the 5% significance level, leading us to **fail to reject the null hypothesis** of a unit root. This suggests that the series are non-stationary.

Conversely, the KPSS test returns p-values of 0.0208 for KWEB and 0.0100 for LIT. These are below the 5% threshold, leading us to **reject the null hypothesis** of stationarity around a deterministic trend.

Since the ADF and KPSS tests point in the same direction, supporting non-stationarity, we can confidently conclude that the log-price series of both ETFs are non-stationary in level. This justifies the use of first differencing, as performed in the ARIMA(6,1,0) models in the previous section.

4.3 Engle-Granger Cointegration Test

To assess whether two non-stationary time series share a common long-term equilibrium relationship, we use the Engle–Granger cointegration test. Cointegration implies that although the individual series may drift over time, a linear combination of them remains stationary, suggesting a stable, mean-reverting relationship in the long run.

This concept is particularly relevant in finance for identifying assets or indices that move together over time, despite short-term fluctuations. Detecting cointegration can support strategies such as pairs trading, where temporary divergences from equilibrium are exploited.

The Engle–Granger procedure consists of estimating a static regression between the two series, then testing the residuals for stationarity. If the residuals are stationary, the original series are said to be cointegrated.

In our case, we test the cointegration between **KWEB** and **LIT**. The test produces a test statistic of `-1.8547`, with a p-value of `0.6028`. This value is far above the 5% significance threshold, and the test statistic is well above the critical value of `-3.3479` at the 5% level. We therefore **fail to reject the null hypothesis** of no cointegration.

The two series are not cointegrated. This suggests that KWEB and LIT do not share a stable long-term relationship, despite both being non-stationary individually. Their price paths may diverge over time without any natural tendency to return to a common equilibrium.

5 Conclusion

This empirical project analyzed the weekly log-prices and returns of three sector-specific ETFs: **KWEB** (Chinese internet and technology), **LIT** (lithium and battery technology), and **URA** (uranium and nuclear energy), over the period from January 2015 to January 2025.

The study began with a normalization of price series, highlighting substantial performance differences across sectors. Descriptive statistics revealed that **LIT** had the highest average weekly return but also exhibited the most pronounced tail risk and negative skewness, suggesting a greater likelihood of extreme negative events. All three ETFs displayed excess kurtosis, indicating fat tails and departures from normality.

Normality tests, correlation matrices, and autocorrelation functions confirmed the presence of non-Gaussian and weakly dependent structures, justifying further econometric modeling. ARIMA(6,1,0) models were applied to the log-prices of **KWEB** and **LIT**, with none of the autoregressive terms showing statistical significance at the 5% level. Impulse response functions showed that shocks had limited persistence, decaying rapidly over time.

Unit root testing via ADF and KPSS confirmed that both log-price series were non-stationary, validating the use of differencing in the ARIMA framework. Finally, the Engle–Granger test for cointegration between **KWEB** and **LIT** indicated the absence of a long-term equilibrium relationship, suggesting independent long-run behavior between the two ETFs.

In summary, the ETFs exhibit low short-term predictability and no evidence of long-term co-movement, consistent with weak-form market efficiency. This study underscores the importance of combining time series diagnostics with financial insight to understand the dynamic behavior of asset prices.

Future extensions could include multivariate models such as VAR or VECM, high-frequency data analysis, or structural break detection to refine the understanding of regime shifts and macroeconomic influences in ETF markets.